

Supplement D — Fullrace Replication

fullrace_granularity_sweep.csv

16 rows x 14 columns.

Columns: mcs, effective_min_samples, min_samples_passed, n_clusters, noise_fraction, assigned_fraction, DBCV, silhouette, min_cluster_size_observed, median_cluster_size, max_cluster_size, mean_enriched_terms, frac_interpretable, degenerate_flag

fullrace_primary_solution.csv

48 rows x 9 columns.

Columns: fr_cluster, size, n_enriched, top_terms, races_present, mean_cosine_to_centroid, representative_1_sha1, representative_2_sha1, representative_3_sha1

fullrace_seed_stability.csv

3 rows x 16 columns.

Columns: mcs, n_seeds, viable_seeds, failed_seeds, degenerate_seeds, n_clusters_ref, n_clusters_min, n_clusters_median, n_clusters_max, noise_min, noise_median, noise_max, ari_median, ari_min, hungarian_median, hungarian_min

fullrace_cluster_enrichment.csv

1374 rows x 7 columns.

Columns: fr_cluster, term, in_cluster, cluster_size, p_value, significant, weight

cross_register_correspondence.csv

34 rows x 18 columns.

Columns: highlight_cluster, highlight_name, domain, availability_class, matched10_sentences, best_fr_cluster, fr_size, fr_races_present, centroid_cosine, lexical_weighted_jaccard, lexical_overlap_at_15, null_p95, null_p_empirical, exceeds_null, n_fr_above_threshold, fr_clusters_above_threshold, relation, interpretable_non_recovery

cross_register_null.csv

34 rows x 10 columns.

Columns: highlight_cluster, highlight_name, availability_class, observed_best_cosine, null_median, null_p95, null_p_empirical, exceeds_null_p95, within_highlight_offdiag_p95, exceeds_within_highlight_p95

matched_event_availability.csv

34 rows x 8 columns.

Columns: highlight_cluster, highlight_name, domain, canonical_total_sentences, matched10_sentences, matched10_events, share_of_canonical_total, availability_class

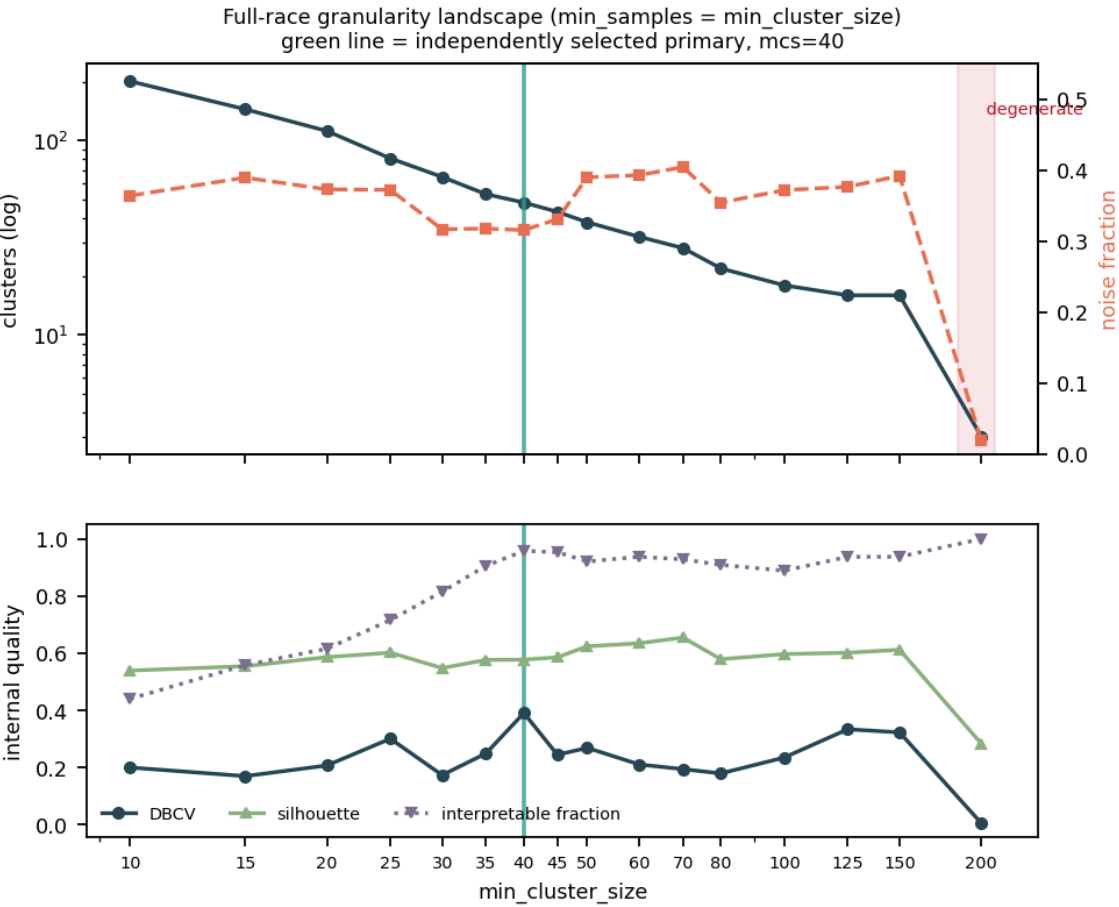
highlight__cluster	highlight_name	domain	canonical__total__sentences	matched10__sentences	matched10__events	share_of__canonical__total	availability__class
0	DRS-assisted battles	Race dynamics	205	10	5	0.0487805	SPARSE
1	Time penalties	Regulatory	118	11	4	0.0932203	SPARSE
2	Lights-out race starts	Race dynamics	54	1	1	0.0185185	EFFECTIVELY__ABSENT
3	Mirror checks defending	Race dynamics	47	6	6	0.12766	SPARSE
4	Race start launch	Race dynamics	42	4	4	0.0952381	EFFECTIVELY__ABSENT
5	Weather conditions	Strategy & technical	43	3	3	0.0697674	EFFECTIVELY__ABSENT
6	Grand Prix wins	Narrative & meta	465	55	10	0.11828	WELL__REPRESENTED
7	Race start lights	Race dynamics	36	3	3	0.0833333	EFFECTIVELY__ABSENT
8	Race control flags	Regulatory	118	11	6	0.0932203	SPARSE
9	Formula 1 milestones	Narrative & meta	134	12	8	0.0895522	SPARSE
10	Chicane moves	Race dynamics	62	5	2	0.0806452	SPARSE
11	Team radio celebrations	Narrative & meta	55	4	3	0.0727273	EFFECTIVELY__ABSENT
12	Podium finishes	Narrative & meta	139	8	5	0.057554	SPARSE
13	Championship & title	Narrative & meta	158	18	8	0.113924	WELL__REPRESENTED
14	Pit stops	Strategy & technical	185	17	8	0.0918919	WELL__REPRESENTED
15	Tires & compounds	Strategy & technical	379	32	8	0.0844327	WELL__REPRESENTED
16	Safety car deployment	Regulatory	122	12	6	0.0983607	SPARSE
17	Gap & pace	Strategy & technical	67	6	3	0.0895522	SPARSE
18	Team dynamics	Narrative & meta	600	67	10	0.111667	WELL__REPRESENTED
19	Race lead holding	Race dynamics	115	8	7	0.0695652	SPARSE
20	Wing & car damage	Strategy & technical	61	4	3	0.0655738	EFFECTIVELY__ABSENT
21	Braking & lockups	Race dynamics	128	13	9	0.101562	SPARSE
22	Wheel-to-wheel	Race dynamics	71	7	5	0.0985915	SPARSE
23	Circuit action	Race dynamics	72	10	7	0.138889	SPARSE
24	Inter-car distance	Strategy & technical	60	1	1	0.0166667	EFFECTIVELY__ABSENT
25	Driver & team references	Narrative & meta	69	13	9	0.188406	SPARSE
26	Defending & attacking	Race dynamics	110	10	6	0.0909091	SPARSE
27	Overtake battles	Race dynamics	762	68	10	0.0892388	WELL__REPRESENTED
28	Corner battles	Race dynamics	82	4	3	0.0487805	EFFECTIVELY__ABSENT
29	Corner specific moves	Race dynamics	60	8	3	0.133333	SPARSE
30	Lap pace & timing	Strategy & technical	136	14	7	0.102941	SPARSE
31	Race leader changes	Race dynamics	162	13	6	0.0802469	SPARSE
32	Front position changes	Race dynamics	45	2	2	0.0444444	EFFECTIVELY__ABSENT
33	Field position changes	Race dynamics	122	9	5	0.0737705	SPARSE

semantic_lexical_correspondence.csv

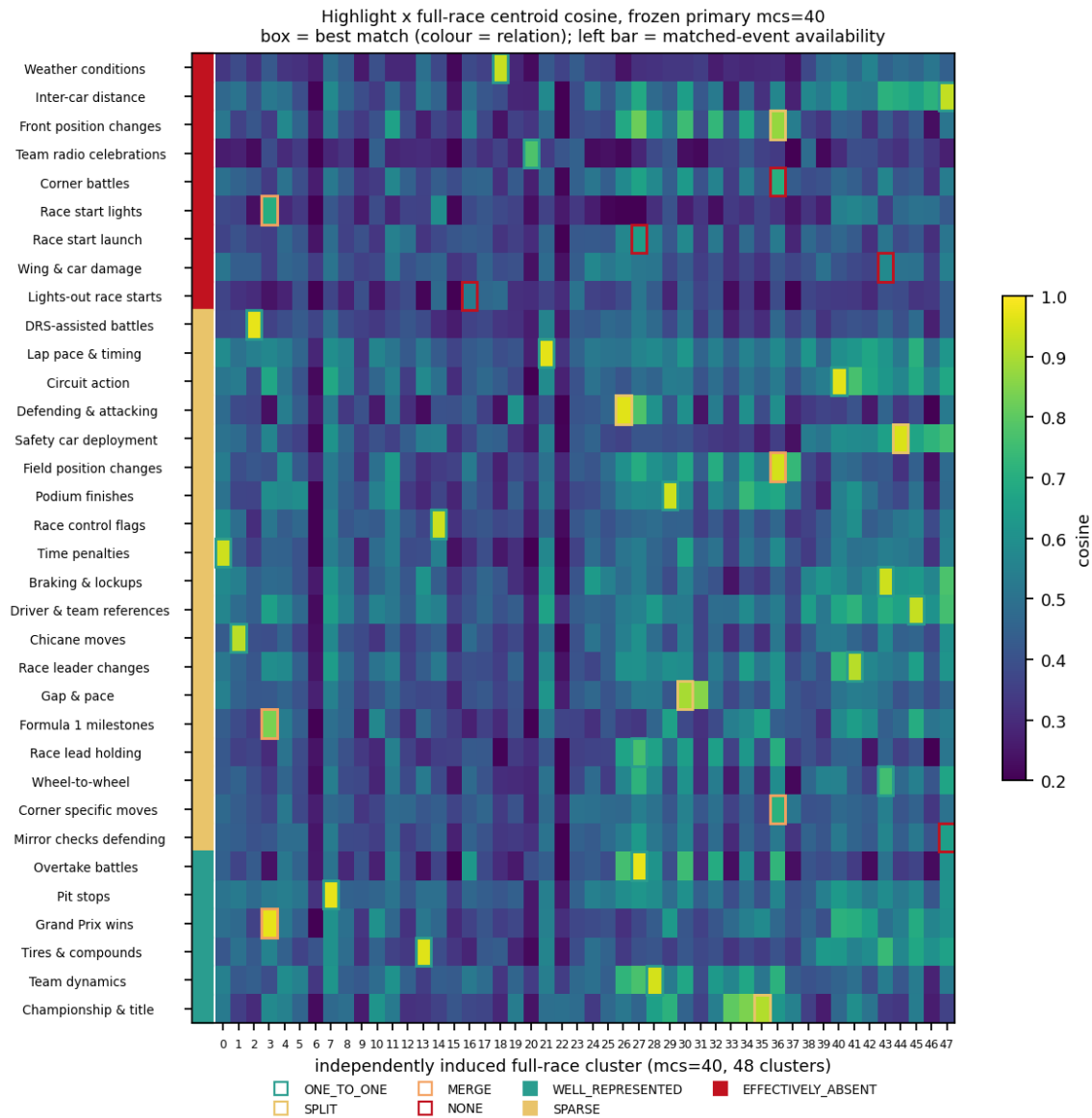
34 rows x 49 columns.

Columns: Unnamed: 0, F0, F1, F2, F3, F4, F5, F6, F7, F8, F9, F10, F11, F12, F13, F14, F15, F16, F17, F18, F19, F20, F21, F22, F23, F24, F25, F26, F27, F28, F29, F30, F31, F32, F33, F34, F35, F36, F37, F38, F39, F40, F41, F42, F43, F44, F45, F46, F47

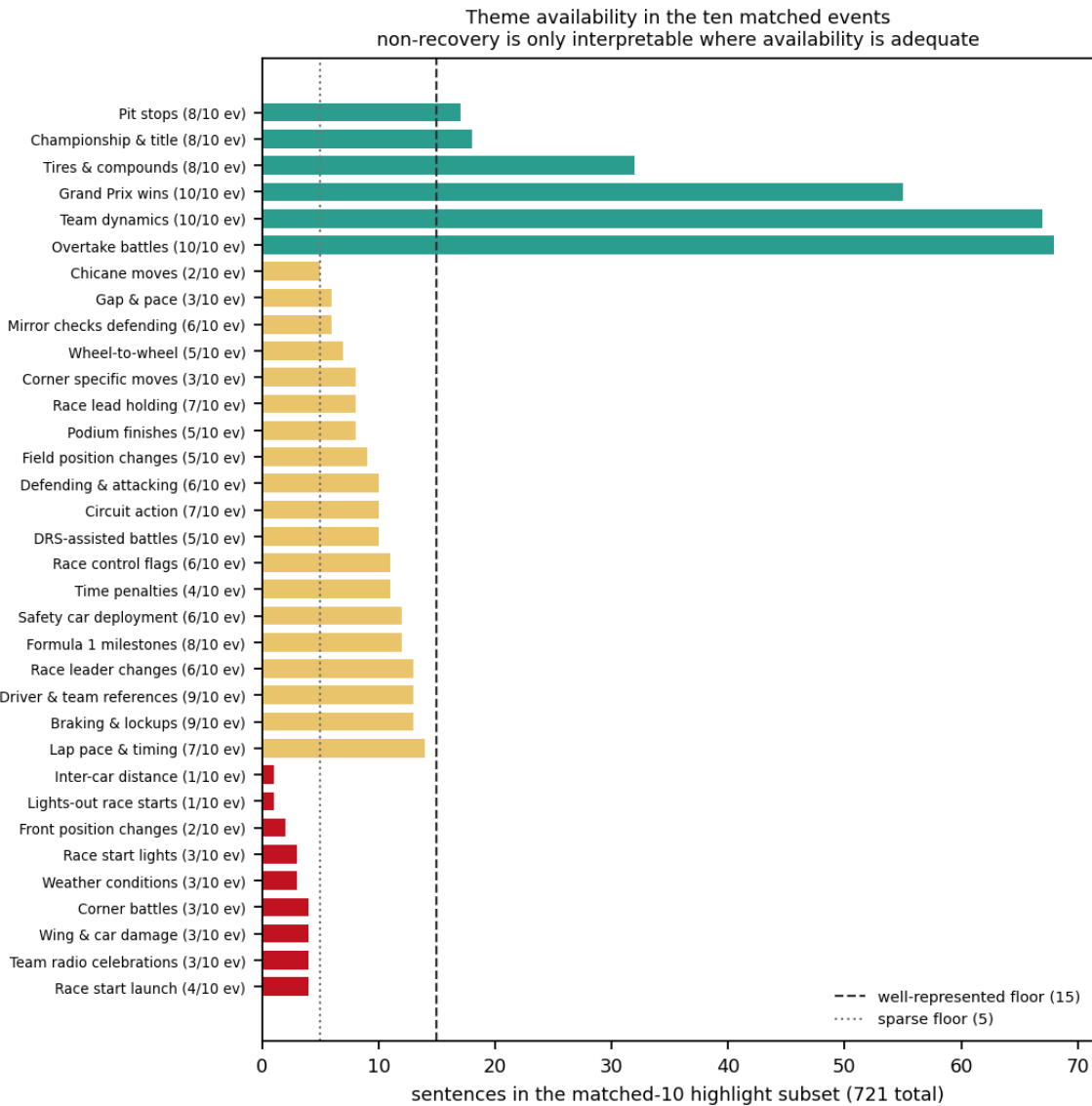
Figures



Full-race granularity landscape; the selected setting is marked.



Highlight x full-race centroid cosine with correspondence type.



Theme availability in the ten matched events.